Table of Contents

1. High Speed Flight – Why Mach Number Matters
2. Speed of Sound
3. Machmeter Principle of Operation
4. Machmeter Construction
5. Machmeter Errors
6. Blockages
7. Abbreviations
8. Climb at Constant CAS – Standard Atmosphere
9. Descent at Constant Mach Number – Standard Conditions
10. Climb/Descent through Isothermal Layer and Inversion
11. Climb/Descent Summary
12. Worked Example Problems
13. Mach/Airspeed Indicator
14. Practice Questions & Detailed Answers
15. Master Reference Tables

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1. High Speed Flight – Why Mach Number Matters

> **What this section covers:** The dangers of high-speed flight and the need for a Machmeter.

As an aircraft approaches the **local speed of sound**, airflow over parts of the fuselage or wings may be accelerated to, or beyond, the speed of sound, forming **shock waves**. Consequences include:

- Increased drag
- Reduced lift
- **Mach tuck** — a sudden, severe downward-pitching trim change
- Buffeting
- Reduction in, or loss of, control effectiveness

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A **limiting Mach number (M_{MO})** is specified for each aircraft based on flight trials. This must not be exceeded. The Machmeter displays the current Mach number so the pilot can remain below M_{MO}.

> **M_{MO} — Never Exceed Mach Number:** Determined by flight trials. Exceeding it risks Mach tuck, buffet, and loss of control. The Machmeter provides the primary speed reference at altitude.

2. Speed of Sound

> **What this section covers:** The formula for Local Speed of Sound (LSS) and how it varies with temperature and altitude.

The speed of sound is **not constant** — it varies with **air temperature** only, not pressure or density directly.

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```
LSS = 38.95 × √T

Where:
  LSS = Local Speed of Sound (knots)
  38.95 = constant
  T = Absolute temperature (Kelvin)
  (0°C = 273K; T(K) = T(°C) + 273)
```

Key Values — memorise:

ISA MSL (+15°C = 288K): LSS = 38.95 × √288 = **661 knots**
FL300 (−45°C = 228K): LSS = 38.95 × √228 = **589 knots**
Tropopause and above (−56.5°C = 216.5K): LSS = 38.95 × √216.5 ≈ **573 knots**

Key Rule: Higher temperature → higher LSS. Temperature normally falls with altitude → LSS normally falls with altitude. Above the tropopause (ISA) temperature is constant → LSS is constant at ~573 kt.

```
graph LR
    A["Temperature Increases"] --> B["LSS Increases"]
    C["Altitude Increases (below tropopause)"] --> D["Temperature Decreases"] --> E["LSS Decreases"]
    F["Above Tropopause (ISA)"] --> G["Temperature Constant −56.5°C"] --> H["LSS Constant ~573 kt"]
```

3. Machmeter Principle of Operation

What this section covers: How the Machmeter derives Mach number from pitot and static pressures.

```
Mach Number (MN) = TAS / LSS

Since MN ∝ √(Dynamic Pressure / Static Pressure)
         = √((Pitot Pressure − Static Pressure) / Static Pressure)

Density (ρ) cancels → MN is proportional to √(Dp / Ps)
where Dp = dynamic pressure, Ps = static pressure
```

The Machmeter uses **two capsules**:

1. **Airspeed Capsule** — expands/contracts with dynamic pressure (pitot − static). Senses TAS component.
2. **Altitude (Aneroid) Capsule** — expands/contracts with changes in static pressure in the instrument case. Compensates for the effect of altitude on the speed of sound.

Their combined mechanical output via the **ratio arm** and **ranging arm** drives the pointer to display Mach number directly.

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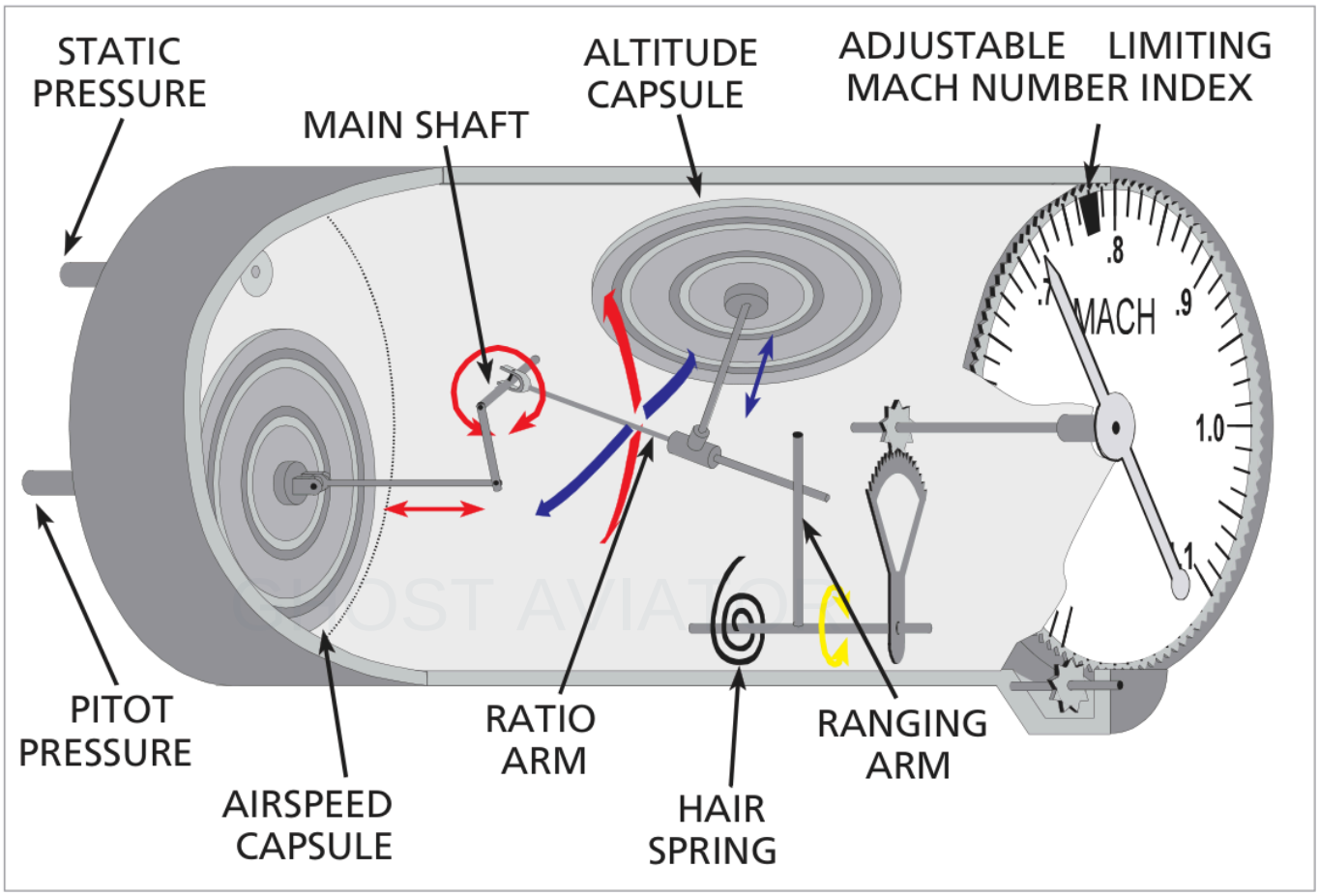


Fig. 7.1 – The Machmeter: airspeed capsule, altitude capsule, ratio arm, and ranging arm —

4. Machmeter Construction

What this section covers: How the two capsules interact to produce a Mach number reading.

- **Static pressure** enters the instrument case (acts on the altitude capsule).
- **Pitot pressure** is fed directly into the airspeed capsule.
- Expansion of the airspeed capsule is transmitted via the **airspeed link and main shaft** to the ratio arm.
- The altitude capsule also governs the position of the ratio arm.
- The **spring-loaded ranging arm** transmits ratio arm movement to the pointer.

Rule: If either or both capsules *expand* (IAS increases and/or altitude increases), the ranging arm rotates and indicated Mach number **increases**. Reverse for a decrease.

An **adjustable Mach limit index** can be set via a small knob to the aircraft's M_{MO}, providing a visual speed warning.

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5. Machmeter Errors

What this section covers: Which errors affect the Machmeter and which do not — a frequent exam focus.

Errors the Machmeter DOES suffer:

- **Instrument Error** — manufacturing imperfections
- **Position Error** — disturbed airflow at pitot/static sources; can change sign at high Mach numbers
- **Manoeuvre-Induced Error** — unpredictable changes in static source airflow during manoeuvres

Errors the Machmeter does NOT suffer:

- **Temperature Error** — cancels out (density appears in both numerator and denominator)
- **Density Error** — same reason, density cancels
- **Compressibility Error** — the Machmeter is calibrated to the dynamic/static pressure ratio, so compressibility is inherently calibrated out

5.1 Position Error at High Mach Numbers

At higher Mach numbers, changes in airflow can cause position error to increase in magnitude and even **change sign**. If position error causes the Machmeter to under-read, this could become dangerous. The normal arrangement in modern jet transport aircraft is to bias instrument and position error so the Machmeter **always over-reads** — providing a safety margin.

Exam Tip: The Machmeter is NOT affected by temperature or density errors. Only three errors: instrument, position, manoeuvre-induced. This contrasts with the ASI which also has density and compressibility errors.

6. Blockages

What this section covers: Effect of pitot and static blockages on Machmeter readings — note these are identical to ASI blockage effects.

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Source Blocked	Condition	Effect on Machmeter
Static	Climb at constant IAS	Altitude capsule doesn't move; airspeed capsule contracts (static in pitot falls but case static frozen) → **Under-reads**
Static	Descent at constant IAS	Airspeed capsule expands (pitot static rises, case static frozen) → **Over-reads**
Pitot	Climb at constant IAS	Airspeed capsule expands in error (capsule static > case static) → **Over-reads**
Pitot	Descent at constant IAS	Airspeed capsule contracts (capsule static < case static) → **Under-reads**

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> **Exam Tip:** Machmeter blockage errors are identical to ASI blockage errors. Static blocked in climb → under-reads; static blocked in descent → over-reads. Pitot blocked is the opposite.

7. Abbreviations

Abbreviation	Meaning
MMR	Machmeter Reading — the uncorrected raw reading
IMN	Indicated Mach Number — MMR corrected for instrument error (values quoted in Flight Manuals are normally IMN)
TMN	True Mach Number — IMN corrected for position error
MMO	Maximum Operating Mach Number — the never-exceed limit

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8. Climb at Constant CAS – Standard Atmosphere

> **What this section covers:** How TAS and Mach number change during a constant-CAS climb.

Example: Climbing at `330 kt CAS` from sea level to `FL360` in ISA:

- TAS increases from `330 kt` to `593 kt`
- Mach number increases from `M 0.5` to `M 1.05`

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This rapid rise in Mach number is why high-performance aircraft are flown at **constant CAS (or IAS)** in the lower part of the climb, then transition to a **constant Mach number** for the remainder — to avoid inadvertently exceeding M_{MO}.

9. Descent at Constant Mach Number – Standard Conditions

> **What this section covers:** How TAS and CAS change during a constant Mach descent.

During a descent in ISA, temperature *increases* → LSS increases. To maintain constant Mach number, TAS must also increase (MN = TAS/LSS). As density also increases during descent, CAS increases even more rapidly (Dynamic Pressure = ½ρV^2). Example: M 0.8 descent from FL400 to MSL:

- At `FL400`: TAS = `450 kt`, CAS = `242 kt`
- At `MSL`: TAS = `528 kt`, CAS = `528 kt` (would exceed V_{MO})

This is why CAS is used in the descent rather than constant Mach number below a certain altitude.

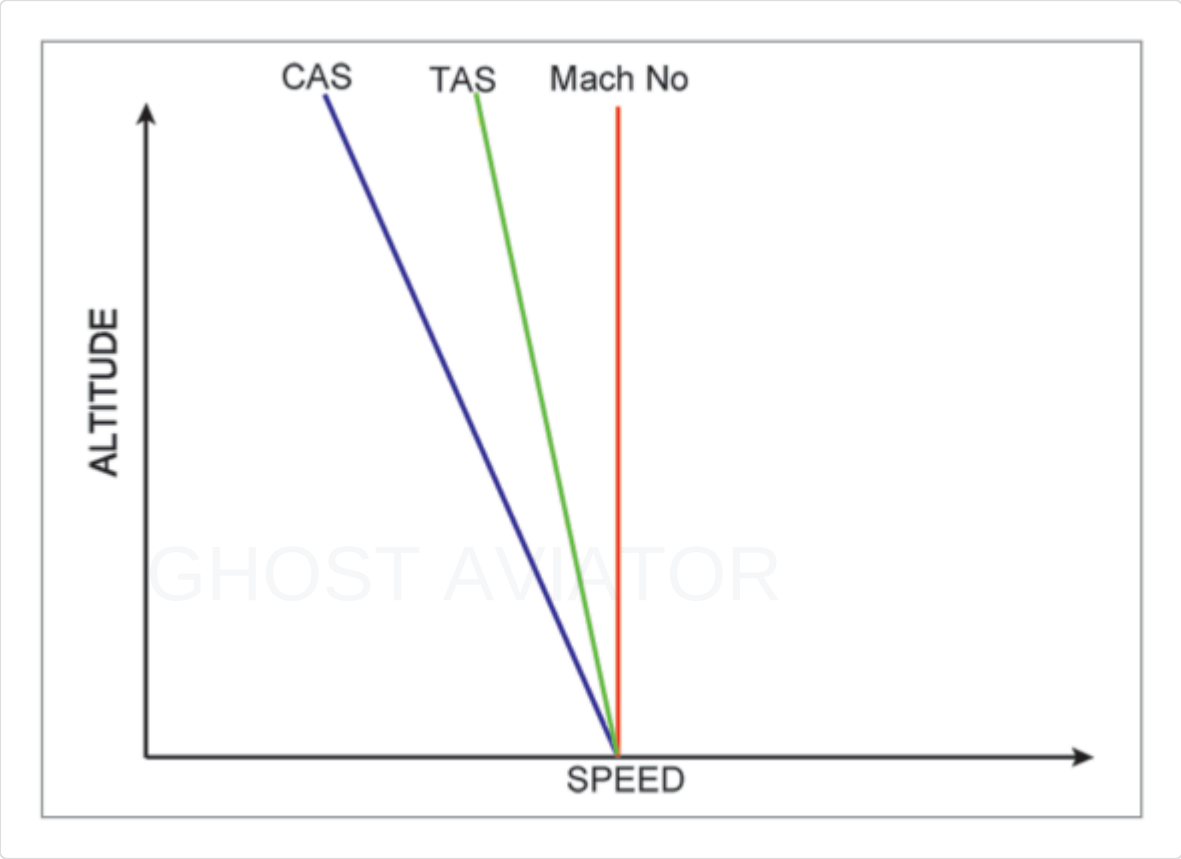


Fig. 7.3 – Constant Mach descent: as altitude decreases, TAS and CAS both increase —

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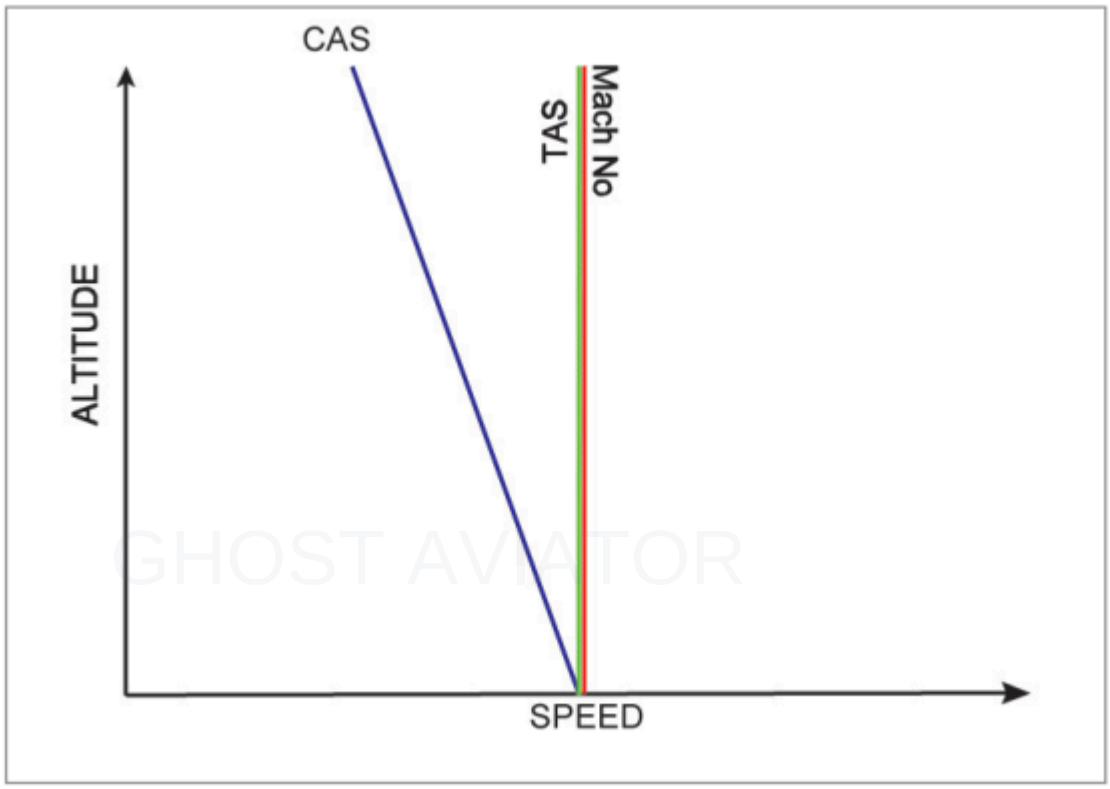
10. Climb/Descent through Isothermal Layer and Inversion

> **What this section covers:** How temperature anomalies affect the CAS/TAS/Mach relationship.

10.1 Isothermal Layer (temperature constant)

Condition	LSS	TAS	CAS
Constant Mach — climbing	Constant (temp constant)	Constant (MN×LSS)	Decreases (density reduces)
Constant Mach — descending	Constant	Constant	Increases (density increases)
Constant CAS — climbing	Constant	Increases	Constant

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Fig. 7.4 – Isothermal layer: TAS constant at constant Mach, CAS changes with density —

10.2 Inversion (temperature increases with altitude)

Condition	LSS	TAS	CAS
Constant Mach — climbing	Increases (temp rises)	Increases (MN×LSS)	Decreases (density reduces faster than TAS rises)
Constant Mach — descending	Decreases	Decreases	Increases
Constant CAS — climbing	Increases	Increases (greater rate than MN)	Constant

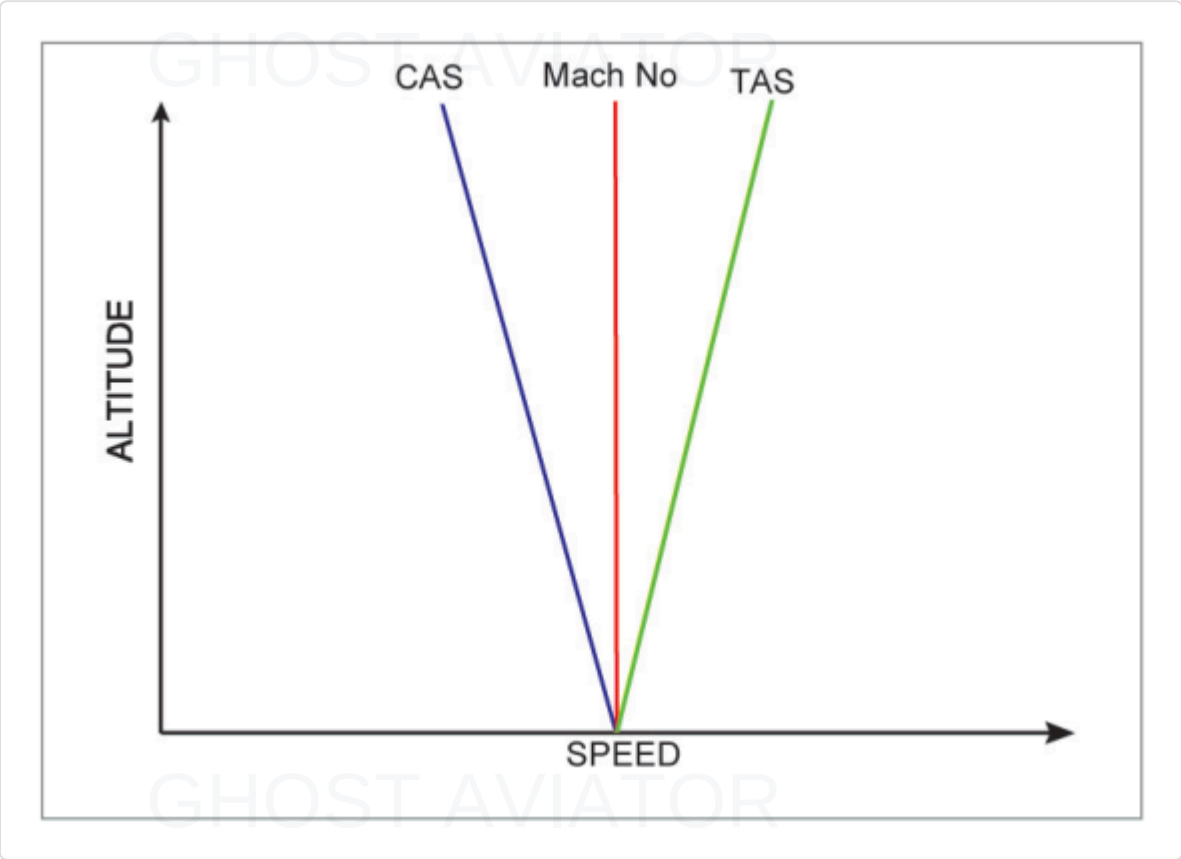


Fig. 7.5 – Inversion: TAS increases at constant Mach because LSS increases with temperature —

11. Climb/Descent Summary

Universal Rules (apply in all conditions):

- TAS **always increases** when climbing at a constant CAS.
- Climbing at constant TAS → CAS **always decreases**.
- Climbing at constant CAS → Mach number **always increases**.
- Climbing at constant Mach → CAS **always decreases**.

Reason: Pressure effect on air density dominates over temperature variation in LSS.

```
flowchart TD
    A["Climb"] --> B{"What is constant?"}
    B -->|CAS| C["TAS increases
Mach number increases"]
    B -->|TAS| D["CAS decreases
Mach number: depends on temp"]
    B -->|Mach number| E["TAS depends on LSS
CAS always decreases"]
```

Quick Revision – Sections 8–11: Constant CAS climb → TAS↑ Mach↑. Constant Mach climb → CAS↓ TAS depends on temp. The climb/descent diagrams (Fig 7.2 & 7.3) are mirror images. In isothermal: constant Mach = constant TAS. In inversion: constant Mach = increasing TAS (climb).

12. Worked Example Problems

What this section covers: Five representative Machmeter calculation problems from the source text.

Problem 1: Speed of Sound at FL380 (ISA)

FL380 is above the tropopause → Temperature = −56.5°C = 216.5K
LSS = 38.95 × √216.5 = 38.95 × 14.71 = **573 knots**

Problem 2: TAS at M 0.70, JSA MSL (+15°C = 288K)

LSS = 38.95 × √288 = 38.95 × 16.97 = 661 kt
TAS = MN × LSS = 0.70 × 661 = **463 knots**

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Problem 3: Altitude at which TAS 450 kt = Mach 0.80 (JSA)

LSS = TAS / MN = 450 / 0.8 = 562.5 kt
√T = 562.5 / 38.95 = 14.44 → T = 14.44^2 = 209K = −64°C
−64°C occurs at **FL395** in the JSA (no tropopause in JSA).

Problem 4: LSS from Mach and TAS change

ΔMN = 0.12, ΔTAS = 80 kt
LSS = ΔTAS / ΔMN = 80 / 0.12 = **667 knots**

Problem 5: Temperature deviation analysis (isothermal layer)

At FL360, JSA temp = −57°C. JSA +9 = −48°C.
Same Mach (0.8) and TAS (467 kt) at FL320 → LSS unchanged → temperature unchanged = −48°C.
JSA at FL320 = −49°C → deviation = **+1°C**.
(This shows the aircraft is flying through an isothermal layer.)

Navigation Computer Tips:

- Set Mach index arrow against temperature (°C) in Airspeed window.
- Read TAS against Mach number on inner scale.
- M 1.0 is marked as the blue "10" on the inner scale.

13. Mach/Airspeed Indicator

What this section covers: The combined Mach/ASI instrument and its two types.

Since commercial aircraft need both IAS and Mach indications, the instruments are combined. Two versions exist:

1. **Self-contained instrument** — fed directly from pitot and static sources.
2. **ADC-fed instrument** — receives computed data from the Air Data Computer.

Construction Features

- Airspeed pointer moves **clockwise** over a **fixed scale**.
- From M 0.5, Mach number is read from the same pointer against a **moving Mach scale** that rotates **anticlockwise** as Mach increases.
- A striped VMO needle may mark the maximum operating airspeed.

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ADC-fed combined instrument advantages:

- Corrects for instrument and position errors → displays **CAS** instead of IAS.
- Can show digital displays for both Mach and CAS.

Errors of the Combined Instrument: Has the errors of BOTH the Machmeter AND the ASI: instrument, position, manoeuvre-induced, density, and compressibility errors.

Quick Revision – Chapter 7:

- LSS = 38.95 × √T (Kelvin) in knots. At MSL ISA = 661 kt; tropopause = 573 kt.
- Machmeter uses 2 capsules: airspeed + altitude (aneroid).
- Errors: instrument, position, manoeuvre-induced ONLY. No temp/density/compressibility error.
- Static blocked in climb → under-reads; in descent → over-reads (same as ASI).
- Constant CAS climb: TAS↑ Mach↑. Constant Mach climb: CAS↓.
- Isothermal: constant Mach = constant TAS. Inversion: constant Mach = increasing TAS (climb).

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Practice Questions & Detailed Answers

Questions 1–9 with answer key.

Q1. The local speed of sound is equal to: (K = Constant)

a. K √ temperature (°F) knots

b. K √ temperature (K) knots

c. K √ temperature (°C) knots

d. K √ temperature (K) metres per second

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Correct Answer: **(b)** K √ temperature (K) knots

Explanation: The formula LSS = 38.95 × √T requires T in **Kelvin (absolute temperature)** and gives LSS in **knots**. Using °C or °F would give incorrect results because negative Celsius values would make the square root imaginary. See Section 2.

Why the others are wrong:

- **(a)** °F cannot be used — the formula requires absolute temperature.
- **(c)** °C cannot be used because 0°C ≠ 0 absolute; must add 273 to convert to Kelvin.
- **(d)** The unit is knots, not m/s (a different constant would be required for m/s).

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Instructor's Note: *Always convert to Kelvin: T(K) = T(°C) + 273. This is non-negotiable for all LSS calculations.*

Q2. At FL350 with a JSA deviation of −12, the true airspeed when flying at M 0.78 is:

a. 460 kt

b. 436 kt

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c. 447 kt

d. 490 kt

Correct Answer: **(b)** 436 kt

Explanation: JSA FL350 temperature = −70°C (JSA lapse 2°/1000ft, 35×2=70, MSL +15 −70 = −55°C... wait, let me recalculate: JSA at FL350 = 15 − (35×2) = 15 − 70 = −55°C. Deviation −12 → actual temp = −55 − 12 = −67°C = 206K. LSS = 38.95×√206 = 38.95×14.35 = 559 kt. TAS = 0.78×559 ≈ 436 kt. See Section 2 and Section 12.

Why the others are wrong:

- **(a) 460 kt** would correspond to a warmer temperature (ISA or less cold).
- **(c) 447 kt** and **(d) 490 kt** result from incorrect temperature calculations or wrong JSA lapse rate assumptions.

Instructor's Note: *JSA lapse rate is 2°C per 1000 ft with no tropopause (unlike ISA which stops at FL360). Always state whether you're using ISA or JSA in exam solutions.*

Q3. When climbing at a constant Mach number below the tropopause through an inversion:

a. the CAS and TAS will both increase

b. the CAS and TAS will both decrease

c. the CAS will decrease and the TAS will increase

d. the CAS will increase and the TAS will decrease

Correct Answer: **(c)** the CAS will decrease and the TAS will increase

Explanation: In an inversion, temperature rises with altitude → LSS increases. For constant Mach number, TAS = MN × LSS → TAS must increase. Despite TAS increasing, air density decreases faster during the climb, so CAS (which is density-dependent) decreases. See Section 10.2.

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Why the others are wrong:

- **(a)** CAS cannot increase — density effect always dominates in a climb at constant Mach.
- **(b)** TAS cannot decrease — the rising LSS in an inversion forces TAS upward at constant Mach.
- **(d)** Reverses the correct answer.

Instructor's Note: *Constant Mach climb: CAS ALWAYS decreases. TAS behaviour depends on the temperature profile — in an inversion it increases, in ISA it decreases, in isothermal it stays constant.*

Q4. When descending below the tropopause under normal conditions (increasing temperature) at a constant CAS:

a. both TAS and Mach number will decrease

b. both TAS and Mach number will increase

c. the TAS will decrease and the Mach number will increase

d. the TAS will increase and the Mach number will decrease

Correct Answer: **(a)** both TAS and Mach number will decrease

Explanation: This is the mirror image of constant CAS climb. In a constant CAS descent under ISA conditions, density increases → TAS decreases (TAS = CAS × density correction, lower altitude = denser air = lower TAS). LSS increases (higher temperature) but TAS falls faster → Mach number decreases. See Section 8.

Why the others are wrong:

- **(b)** Incorrectly suggests both increase — that would be a constant CAS climb, not descent.
- **(c) & (d)** Incorrect combinations — in a constant CAS descent, the pressure/density effect dominates and both TAS and MN decrease.

Instructor's Note: *Descent at constant CAS is the mirror image of climb at constant CAS. Climb: TAS↑ Mach↑. Descent: TAS↓ Mach↓.*

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Q5. Cruising at FL390, M 0.84 is found to give a TAS of 499 kt. The ISA deviation at this level will be:

a. −17

b. +17

c. +19

d. −19

Correct Answer: **(b)** +17

Explanation: LSS = TAS/MN = 499/0.84 = 594 kt. √T = 594/38.95 = 15.25 → T = 15.25^2 = 232.6K = −40.4°C ≈ −40°C. FL390 is above the ISA tropopause (ISA temp = −56.5°C). Deviation = actual − ISA = −40 − (−56.5) = +16.5 ≈ **+17**. See Section 12.

Why the others are wrong:

- **(a) −17** and **(d) −19** would mean a colder-than-ISA temperature — but we calculated a warmer temperature.
- **(c) +19** is close but the arithmetic gives +17 when rounded correctly.

Instructor's Note: *Process: TAS/MN → LSS → $(LSS/38.95)^2$ → T(K) → T(°C) → compare with ISA. Practice this sequence until automatic.*

Q6. The errors to which the Machmeter is subject are:

a. instrument error, position error, compressibility error and manoeuvre induced error

b. instrument error, position error and manoeuvre induced error

c. instrument error, position error, barometric error, temperature error and manoeuvre induced error

d. instrument error, position error, density error and manoeuvre induced error

Correct Answer: **(b)** instrument error, position error and manoeuvre induced error

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Explanation: The Machmeter is calibrated to the ratio of dynamic to static pressure. Density and temperature cancel from the calculation, so no density or temperature error. Compressibility is inherently calibrated out by design. Only instrument, position, and manoeuvre-induced errors remain. See Section 5.

Why the others are wrong:

- **(a)** Compressibility error does not apply — it's calibrated out in the Machmeter.
- **(c)** No barometric or temperature error in the Machmeter.
- **(d)** No density error — density cancels in the Mach number calculation.

Instructor's Note: *The Machmeter's immunity to temperature, density, and compressibility errors is its key advantage over the ASI. This is a very common exam question.*

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Q7. The relationships between TAS, Mach number (MNo) and local speed of sound (LSS) is:

a. LSS = MNo/TAS

b. MNo = LSS/TAS

c. TAS = MNo × LSS

d. MNo = LSS × TAS

Correct Answer: **(c)** TAS = MNo × LSS

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Explanation: By definition, Mach Number = TAS/LSS. Rearranging: TAS = MNo × LSS. This is the fundamental Mach relationship. See Section 2.

Why the others are wrong:

- **(a)** LSS = MNo/TAS is the inverse of the correct relationship.
- **(b)** MNo = LSS/TAS inverts numerator and denominator.
- **(d)** MNo = LSS × TAS would give huge numbers — dimensionally and physically incorrect.

Instructor's Note: *Start from MN = TAS/LSS and rearrange as needed. This single formula underpins*

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all Mach calculations.

Q8. The Machmeter gives an indication of Mach number by measuring the ratio:

a. pitot pressure / static pressure

b. static pressure / dynamic pressure

c. dynamic pressure / pitot pressure

d. dynamic pressure / static pressure

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Correct Answer: **(d)** dynamic pressure / static pressure

Explanation: Mach number ∝ √(Dynamic Pressure / Static Pressure) where Dynamic Pressure = Pitot − Static. The altitude capsule measures static pressure (denominator), the airspeed capsule measures dynamic pressure (numerator). See Section 3.

Why the others are wrong:

- **(a)** Pitot/static gives the ratio used in ASI calculations (total pressure ratio), not pure Mach.
- **(b)** Inverted ratio — would decrease as Mach increases.
- **(c)** Dynamic/pitot is not a meaningful aerodynamic ratio.

Instructor's Note: *Dynamic pressure = pitot − static. Mach ∝ √(Dp/Ps). The altitude capsule compensates for changing Ps at different altitudes.*

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Q9. An aircraft is flying at FL350 with a JSA deviation of +8. The Mach No. is 0.83 and the TAS 485. If the aircraft descends to FL300 and maintains the same Mach No. and TAS, the JSA deviation will now be:

a. +8

b. −2

c. +2

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d. −18

Correct Answer: **(b)** −2

Explanation: Same Mach (0.83) and same TAS (485) → LSS unchanged → temperature unchanged. From FL350 calculation: LSS = 485/0.83 = 585 kt → T = $(585/38.95)^2$ = 225K = −48°C. JSA FL350 = 15−70 = −55°C; deviation +8 → −55+8 = −47°C ≈ −48°C ✓ (consistent). At FL300: JSA = 15−60 = −45°C. Actual temp is still −48°C → deviation = −48−(−45) = **−3**. Nearest answer is **(b) −2** (rounding differences in calculation). The key insight: same Mach + same TAS = same temperature = isothermal layer, so the deviation changes as standard temperature changes with altitude. See Section 12, Problem 5.

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Why the others are wrong:

- **(a) +8** would mean the deviation is the same — but JSA standard temperature changes between FL350 and FL300.
- **(c) +2** and **(d) −18** result from incorrect temperature or JSA calculations.

Instructor's Note: *Same MN + same TAS at different altitude → same actual temperature → isothermal layer. The JSA deviation changes because the JSA standard temperature changes with altitude while actual temperature stays the same.*

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Master Reference Tables

Parameter	Value	Condition	Section
LSS at MSL ISA (+15°C)	661 knots	T = 288K	§2
LSS at FL300 ISA (−45°C)	589 knots	T = 228K	§2
LSS at tropopause ISA (−56.5°C)	573 knots	T = 216.5K	§2
LSS formula constant	38.95	Output in knots, T in Kelvin	§2
JSA lapse rate	2°C/1000 ft	No tropopause in JSA	§12
Machmeter errors (only)	Instrument, Position, Manoeuvre-induced	No temp/density/compressibility	§5
Static blocked — climb	Under-reads	Constant IAS	§6
Static blocked — descent	Over-reads	Constant IAS	§6

```
KEY FORMULAS:
LSS (kt)     = 38.95 × √T(K)
MN           = TAS / LSS
TAS          = MN × LSS
T(K)         = T(°C) + 273
T(°C)        = T(K) − 273
LSS → T:     T(K) = (LSS / 38.95)²
```

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Mnemonics:

- **Machmeter errors:** "I Put My foot down" → Instrument, Position, Manoeuvre-induced (only 3!)
- **Static blocked climb:** "STATIC CLIMB = UNDER" (same as ASI)
- **Constant CAS climb:** "Everything goes UP" → TAS↑, Mach↑, LSS↓ (exception)
- **Constant Mach climb:** "CAS always DOWN"

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Answer Key

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9
b	b	c	a	b	b	c	d	b

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